PROJECT 2

PROBABILITY & STATISTICAL DATA ANALYSIS (SECI1143-03)

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Introduction

We have use secondary resources healthcare data samples from several states to evaluate their population's health wellbeing. Our goal is to determine whether the health status in these states is within a normal range or if there are deviations that need attention. By analyzing this data, we hope to identify patterns and trends that indicate the overall health condition of these populations. This involves using statistical methods to compare the data against established health benchmarks. Understanding whether the health is in a normal condition can help in making informed decisions about public health policies and resource allocation, ultimately aiming to improve healthcare outcomes globally.

Dataset

The dataset in the Excel file contains comprehensive health-related data for individuals across various states, detailing information such as gender, general health status, days of poor physical and mental health, recent medical checkup history, physical activity, sleep hours, height, weight, BMI, and several health behaviors including alcohol consumption, HIV testing, flu and pneumococcal vaccination, tetanus vaccination status, high-risk health conditions in the past year, and COVID-19 status. Originally, the dataset comprised over 400,000 samples, but we have selected 580 samples to represent global healthcare trends.

From this data, we aim to explore four potential relationships: the correlation between BMI and sleep hours, the relationship between weight and BMI, whether the average BMI in each state is above 30 (the threshold for obesity), and the relationship between gender and HIV testing to determine if these two datasets are dependent or independent.

Data Analysis

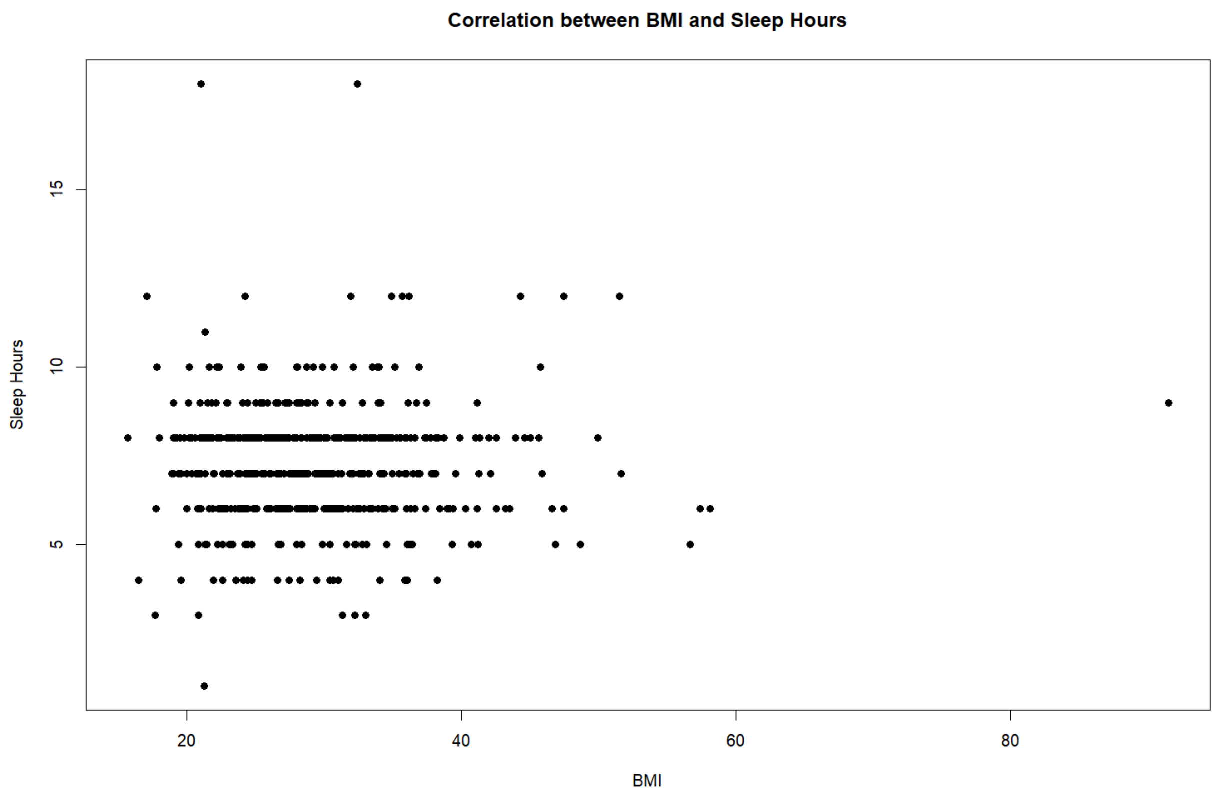
Correlation Analysis

The correlation analysis is used to measure strength of the linear relationship between two variables

Data:

I chose two numerical variables from the healthcare dataset which are sleep hour and BMI. From these data, I want to test the relationship between sleep hour and BMI.

Scatter Plot



Hypothesis

H_0 : There is no relationship between sleep hour and BMI, $\rho = 0$
H_1 : There is a relationship between sleep hour and BMI, $\rho \neq 0$

Test Statistic

Correlation Coefficient, r

```
> print(correlation)
[1] 0.007425639
```

r = 0.0074

$$t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

t = 0.178

d.f. = 577
$\alpha = 0.05$

$t_{a/2} = 1.9641$

$0.178 < 1.9641$

Claim:
Fail to reject H0
There is not enough evidence to support the claim that there is a relationship between sleep hour and BMI.

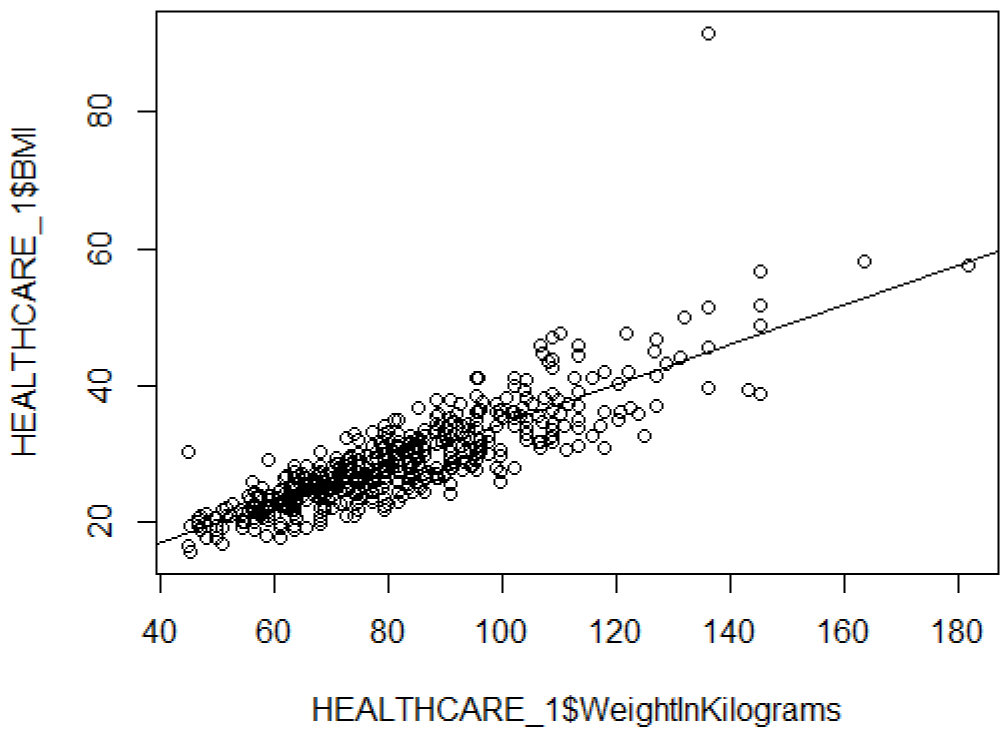
Conclusion:
Since we fail to reject the null hypothesis, we can say that there is no relationship between sleep hour and BMI. Therefore, the amount of sleep an individual gets does not significantly impact their BMI.

Regression test
To check the dependent variable based on the independent variable, we use regression. So, we want to see the impact of change in the independent variable on the dependent variable.

Data
I have chosen two data points from the database Healthcare which is weight and bmi. I want to check whether weight affects bmi or not.

Hypothesis
H_0 : The weight not affects bmi, $\beta_1 = 0$
H_1 : The weight affect bmi, $\beta_1 \neq 0$ (claims)

Test statistic
Scatter plot



Linear regression equation

```
Coefficients:
                    (Intercept)  HEALTHCARE_1$WeightInKilograms
                          5.416                           0.289
```

$b_0 = 5.416$, $b_1 = 0.289$
$y = 5.416 + 0.289x$

Inference of slope
t Test

```
Residuals:
   Min     1Q Median     3Q    Max
-8.853 -2.455 -0.075  1.953 46.810

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                   5.415919   0.663525   8.162 2.07e-15 ***
HEALTHCARE_1$WeightInKilograms 0.288978   0.007852  36.803  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.794 on 577 degrees of freedom
Multiple R-squared:  0.7013,	Adjusted R-squared:  0.7007
F-statistic:  1354 on 1 and 577 DF,  p-value: < 2.2e-16
```

$$\boxed{t = \frac{b_1 - \beta_1}{s_{b_1}}}$$

$$d.f. = n - 2$$

$t = 36.803$

d.f. = 577
$\alpha = 0.05$, $\alpha/2 = 0.025$
$t_{\alpha/2} = 1.960$

$t > t_{\alpha/2}$

Claim:
Fail to accept H_0
There is enough evidence to support the claim that the weight affect bmi

Conclusion:
Since we fail to accept the null hypothesis, we can say that different weight will produce different bmi means that the weight really affects bmi.

Hypothesis 1 Sample Test

A one sample test of means compares the mean of a sample to a pre-specified value and tests for a deviation from that value. We will compare the mean of the data and the mean of the pre-specified value within the sample state.

Data :

I have chosen one data from the database Healthcare which is BMI data. I want to see whether all states compare if this BMI mean is equal to 30.0 which means obesity.

Hypothesis :

H_0: BMI mean is equal to 30.0
H_1: BMI mean is not equal to 30.0

Test Statistic :

Number of sample, $n = 579$
Confident level = 95%
Critical value, $\alpha = 0.05$
Half critical value, $\alpha/2 = 0.025$
Pre-specified mean, $\mu = 30.0$

BMI mean, $\bar{x} = 29.13582$

```
> xdata
[1] 29.13582
```

Standard deviation, $\sigma = 6.936164$

```
> datasd
[1] 6.936164
```

Z statistics

$$Z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Z (calculation) = -2.997947

Z (table on both sides) = (-1.959964, 1.959964)

Z = -2.997947 < -1.959964

Claim:
We reject the null hypothesis, H_0.
Therefore, the mean BMI is not equal to 30.0

Conclusion:
Since we reject the null hypothesis, that means not all the states have the BMI that equals or exceeds to 30.0.

The Chi-square test of independence

The chi-square test of independence checks if two categorical variables are related by comparing observed and expected frequencies in a table. We will compare the x square calculation and x square critical value.

Data :

We have chosen two data points from the database Healthcare, which are gender and HIV testing. We want to compare whether gender and HIV testing are independent or dependent.

Hypothesis :

H_0 : HIV test are independent of gender
H_1 : HIV test are not independent of gender

Test statistic :

Table data:

	No		Yes	
	Observed	Expected	Observed	Expected
Male	325	319.418	66	71.58204
Female	148	153.562	40	34.41796

Degree of freedom = 1
Confident level = 95%
Critical value = 0.05

```
> print("Observed frequency")
[1] "Observed frequency"
> output$observed

        No Yes
Female 325  66
Male   148  40
```

```
> print("Expected frequency")
[1] "Expected frequency"
> output$expected

            No      Yes
Female 319.418 71.58204
Male   153.582 34.41796
```

X square:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

```
> # perform chi-square test on the data table
> chisq.test(tbl,correct=FALSE)

        Pearson's Chi-squared test

data:  tbl
X-squared = 1.641, df = 1, p-value = 0.2002
```

X^2 Calculation = 1.641
P value = 0.2002

```
> #critical value
> alpha <-0.05
> x2.alpha <-qchisq(alpha, df=1,lower.tail=FALSE)
> x2.alpha
[1] 3.841459
```

X^2 Table (critical value) = 3.841459

x^2 (calculation) < x^2 (table)

Claim :

We fail to reject the null hypothesis
Thus , HIV test is independent of gender

Conclusion :
Since we fail to reject the null hypothesis, that means the gender is not affecting someone to have HIV but it's more to the person itself.

Conclusion

The analyses reveal that weight significantly affects BMI while other relationships, such as relationship between sleep hours and BMI and gender and HIV testing, do not show significant effects. From these tests, we learn the possible trends that are related to healthcare that we can spread awareness about. These findings give useful insights into the factors affecting health and behaviors in the dataset.

Appendix

https://www.kaggle.com/datasets/kamilpytlak/personal-key-indicators-of-heart-disease
2022/heart_2022_no_nans

Video Link
https://youtu.be/rRMUZJgRC9w